微分几何习题解答（草案）*

2020 Autumn

§1 第一章习题解答

1. 注意到这样一个事实: 指定$\mathbb{R}^3$中不共面的四点, 则任何一点P的位置可由P到这四点的距离唯一决定. 开集U总包含不共面的四点, 由此可以将U上的等距变换延拓为$\mathbb{R}^3$上的等距变换, 因此U上的等距变换一定是某个合同变换的限制.

2. (1). 由

$$\frac{d}{dt}|\boldsymbol{a}(t)|^2 = \frac{d}{dt}(\boldsymbol{a}(t)\cdot\boldsymbol{a}(t)) = 2\boldsymbol{a}'(t)\cdot\boldsymbol{a}(t),$$

$|\boldsymbol{a}(t)|^2$是常数, 当且仅当$\boldsymbol{a}'(t)\cdot\boldsymbol{a}(t)=0$.

(2). (追加条件: $\boldsymbol{a}(t)$没有零点)

若$\boldsymbol{a}(t)$的方向不变, 则有一个固定的单位向量$\boldsymbol{e}$使得$\boldsymbol{a}(t)=f(t)\boldsymbol{e}$, 其中$f(t)=\boldsymbol{a}(t)\cdot\boldsymbol{e}$处处非零, 故

$$\boldsymbol{a}'(t)=f'(t)\boldsymbol{e}, \qquad \boldsymbol{a}'(t)\times\boldsymbol{a}(t)=\boldsymbol{0}.$$

反之, 若$\boldsymbol{a}'(t)\times\boldsymbol{a}(t)=\boldsymbol{0}$, 则$\boldsymbol{e}(t)=\dfrac{\boldsymbol{a}(t)}{|\boldsymbol{a}(t)|}=\dfrac{\boldsymbol{a}(t)}{f(t)}$是单位向量, 其中$f(t)=|\boldsymbol{a}(t)|$处处非零. 下面只要证明$\boldsymbol{e}(t)$是常向量. 由上一问的结果, $\boldsymbol{e}'(t)\cdot\boldsymbol{e}(t)=0$, 即$\boldsymbol{e}'(t)\cdot\boldsymbol{a}(t)=0$. 而

$$\begin{aligned}\boldsymbol{a}(t) &= f(t)\boldsymbol{e}(t),\\ \boldsymbol{a}'(t) &= f'(t)\boldsymbol{e}(t)+f(t)\boldsymbol{e}'(t),\\ \boldsymbol{a}'(t)\times\boldsymbol{a}(t) &= f(t)\boldsymbol{e}'(t)\times\boldsymbol{a}(t)=\boldsymbol{0},\end{aligned}$$

因此$\boldsymbol{e}'(t)\times\boldsymbol{a}(t)=\boldsymbol{0}$. 令$\boldsymbol{e}'(t)=\lambda(t)\boldsymbol{a}(t)$,

$$\boldsymbol{e}'(t)\cdot\boldsymbol{a}(t)=\lambda(t)\boldsymbol{a}(t)\cdot\boldsymbol{a}(t)=\lambda(t)f^2(t)=0,$$

于是$\lambda(t)=0$, 即$\boldsymbol{e}'(t)=0$, $\boldsymbol{e}(t)$是常向量.

注: 当$\boldsymbol{a}(t)$有零点时, 情况会有变化. 考虑C^∞(非正则)曲线

$$\boldsymbol{a}(t)=\begin{cases}(e^{-1/t^2}\sin\dfrac{1}{t},0,0), & t>0\\ (0,0,0), & t=0\\ (0,e^{-1/t^2}\sin\dfrac{1}{t},0), & t<0\end{cases}$$

*彭家贵、陈卿　著《微分几何》，高等教育出版社2002年7月第1版

3. 爱玩的可以用Mathematica跑一下.

4. (1). $\boldsymbol{e}_{\sigma(i)} \cdot \boldsymbol{e}_{\sigma(j)} = \delta_{\sigma(i)\sigma(j)} = \delta_{ij}$.

(2). $\{O; \boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3\}$与$\{O; \boldsymbol{e}_{\sigma(1)}, \boldsymbol{e}_{\sigma(2)}, \boldsymbol{e}_{\sigma(3)}\}$定向相同, 当且仅当$(\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3) = (\boldsymbol{e}_{\sigma(1)}, \boldsymbol{e}_{\sigma(2)}, \boldsymbol{e}_{\sigma(3)})$, 由性质1.1的(3), 当且仅当$\sigma$是偶置换.

5. 首先需要说明一下, 合同变换$\mathcal{T}$是如何作用在向量$\boldsymbol{v}$上的.

对向量$\boldsymbol{v}$, 取两点P, Q使得$\boldsymbol{v} = \overrightarrow{PQ}$, 定义$\mathcal{T}\boldsymbol{v} := \overrightarrow{\mathcal{T}(P)\mathcal{T}(Q)}$. 在这个意义下, $\mathcal{T}$对$\boldsymbol{v}$的作用只与正交变换那部分有关, 而与平移那部分无关.

下面证明: 对任意三阶方阵$\boldsymbol{M}$, 都有$\boldsymbol{M}^T(\boldsymbol{M}\boldsymbol{a} \times \boldsymbol{M}\boldsymbol{b}) = (\det \boldsymbol{M})(\boldsymbol{a} \times \boldsymbol{b})$.

混合积的几何意义是三个向量生成平行六面体的有向体积, 行列式的几何意义是线性变换下体积的伸缩比. 对任意向量$\boldsymbol{c}$,

$$\begin{aligned}(\det \boldsymbol{M})(\boldsymbol{a} \times \boldsymbol{b}) \cdot \boldsymbol{c} &= (\det \boldsymbol{M}) \det(\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}) \\ &= \det(\boldsymbol{M}\boldsymbol{a}, \boldsymbol{M}\boldsymbol{b}, \boldsymbol{M}\boldsymbol{c}) \\ &= (\boldsymbol{M}\boldsymbol{a} \times \boldsymbol{M}\boldsymbol{b}) \cdot \boldsymbol{M}\boldsymbol{c} \\ &= \boldsymbol{M}^T(\boldsymbol{M}\boldsymbol{a} \times \boldsymbol{M}\boldsymbol{b}) \cdot \boldsymbol{c},\end{aligned}$$

因此$\boldsymbol{M}^T(\boldsymbol{M}\boldsymbol{a} \times \boldsymbol{M}\boldsymbol{b}) = (\det \boldsymbol{M})(\boldsymbol{a} \times \boldsymbol{b})$.

当$\boldsymbol{M}$为正交阵时, 等式化为$\boldsymbol{M}\boldsymbol{a} \times \boldsymbol{M}\boldsymbol{b} = (\det \boldsymbol{M})\boldsymbol{M}(\boldsymbol{a} \times \boldsymbol{b})$. 于是

$$\mathcal{T}\boldsymbol{v} \times \mathcal{T}\boldsymbol{w} = (\operatorname{sgn} \mathcal{T})\mathcal{T}(\boldsymbol{v} \times \boldsymbol{w}),$$

其中$\operatorname{sgn} \mathcal{T}$在刚体运动时取$+1$, 在反向刚体运动时取$-1$.

Supplement

Exercise 1 *Show that the distribution laws of the inner product and the exterior product in $\mathbb{R}^3$, that is*

$$\overrightarrow{a} \cdot \left(\overrightarrow{b} + \overrightarrow{c} \right) = \overrightarrow{a} \cdot \overrightarrow{b} + \overrightarrow{a} \cdot \overrightarrow{c}$$

and

$$\overrightarrow{a} \times \left(\overrightarrow{b} + \overrightarrow{c} \right) = \overrightarrow{a} \times \overrightarrow{b} + \overrightarrow{a} \times \overrightarrow{c}.$$

Proof. Denote by $\overrightarrow{d}^{\top}$ and $\overrightarrow{d}^{\perp}$ the projection of $\overrightarrow{d}$ along $\overrightarrow{a}$ and orthogonal component of $\overrightarrow{d}$ w.r.t $\overrightarrow{a}$ respectively. It is easily observed from the graph that

$$\left(\overrightarrow{b} + \overrightarrow{c} \right)^{\top} = \overrightarrow{b}^{\top} + \overrightarrow{c}^{\top} \text{ and } \left(\overrightarrow{b} + \overrightarrow{c} \right)^{\perp} = \overrightarrow{b}^{\perp} + \overrightarrow{c}^{\perp}.$$

WLOG, we may assume $\overrightarrow{a}$ is a unit vector. So we have

$$\begin{aligned}
& \overrightarrow{a} \cdot \left(\overrightarrow{b} + \overrightarrow{c} \right) \\
= \; & \left(\overrightarrow{b} + \overrightarrow{c} \right) \cdot \overrightarrow{a} \\
= \; & \left(\overrightarrow{b} + \overrightarrow{c} \right)^{\top} \cdot \overrightarrow{a} \\
= \; & \frac{\left(\overrightarrow{b} + \overrightarrow{c} \right)^{\top}}{\overrightarrow{a}} \\
= \; & \frac{\overrightarrow{b}^{\top} + \overrightarrow{c}^{\top}}{\overrightarrow{a}} \\
= \; & \frac{\overrightarrow{b}^{\top}}{\overrightarrow{a}} + \frac{\overrightarrow{c}^{\top}}{\overrightarrow{a}} \\
= \; & \overrightarrow{b}^{\top} \cdot \overrightarrow{a} + \overrightarrow{c}^{\top} \cdot \overrightarrow{a} \\
= \; & \overrightarrow{b} \cdot \overrightarrow{a} + \overrightarrow{c} \cdot \overrightarrow{a} \\
= \; & \overrightarrow{a} \cdot \overrightarrow{b} + \overrightarrow{a} \cdot \overrightarrow{c}
\end{aligned}$$

where the quotient of any two parallel vectors means the multiple. This complete the part of the inner product. As for the case of the exterior product, it could be obtained by

$$\begin{aligned}
& \overrightarrow{a} \times \left(\overrightarrow{b} + \overrightarrow{c} \right) \\
= \; & \overrightarrow{a} \times \left(\overrightarrow{b} + \overrightarrow{c} \right)^{\perp} \\
= \; & \overrightarrow{a} \times \left(\overrightarrow{b}^{\perp} + \overrightarrow{c}^{\perp} \right) \\
= \; & \overrightarrow{a} \times \overrightarrow{b}^{\perp} + \overrightarrow{a} \times \overrightarrow{c}^{\perp} \\
= \; & \overrightarrow{a} \times \overrightarrow{b} + \overrightarrow{a} \times \overrightarrow{c}.
\end{aligned}$$

Here, in the third equality, we use the geometric interpretation (more intuitive) that the exterior product with a unit vector is equivalent to the rotation along such unit vector by $\frac{\pi}{2}$ which preserve the addition of vectors. ■